

Defining, assessing, and developing creativity in sport: a systematic narrative review

Francisco de Sa Fardilha & Justine B. Allen

To cite this article: Francisco de Sa Fardilha & Justine B. Allen (2019): Defining, assessing, and developing creativity in sport: a systematic narrative review, International Review of Sport and Exercise Psychology, DOI: [10.1080/1750984X.2019.1616315](https://doi.org/10.1080/1750984X.2019.1616315)

To link to this article: <https://doi.org/10.1080/1750984X.2019.1616315>



Published online: 14 May 2019.



Submit your article to this journal [↗](#)



Article views: 37



View Crossmark data [↗](#)



Defining, assessing, and developing creativity in sport: a systematic narrative review

Francisco de Sa Fardilha  and Justine B. Allen 

Faculty of Health Sciences and Sport, University of Stirling, Stirling, UK

ABSTRACT

Research on creativity in sport is gaining momentum, due to a growing interest from coaches and academics in developing strategies to increase unpredictability in individual and collective behaviour which may allow teams to gain an important advantage over their opponents. The purpose of this paper was to conduct the first systematic narrative review of the literature on sporting creativity, critically synthesising 51 years of published research (1967–2018) and proposing avenues for future research. Six databases were used, and 48 documents met search criteria. The findings are organised in four categories: (a) defining creativity, (b) correlates of creativity, (c) assessing creativity and (d) developing creativity. Creativity definitions and assessments have privileged thought processes over the ability to act. A distinction is warranted between creativity about sport and creativity in sport (in action) and aligned assessment methods. The literature does not support a single strategy for the development of sporting creativity but does support its trainability. Evidence of the effectiveness of programmes for the enhancement of sporting creativity is growing but is still limited. Furthermore, while it is recognised that coaches have a pivotal role in the development of sporting creativity, research involving them is still scarce.

ARTICLE HISTORY

Received 23 September 2018
Accepted 29 April 2019

KEYWORDS

Creativity; sport; complexity;
deliberate practice;
deliberate play

Introduction

Creativity is a topic that attracts attention from all areas of society and domains of performance (Runco, 2014). Dietrich and Haider (2017, p. 1) describe it as the ‘fountainhead of our civilizations and a defining characteristic of what makes us human’. Since J. P. Guilford’s historical speech as part of the American Psychological Association’s (APA) Presidential Address in 1950, research on creativity has risen dramatically, even though not to a level that can reflect ‘its importance both to the field of psychology and to the world’ (Sternberg & Lubart, 1999, p. 12). Over the past seven decades, many different lines of research have been explored – e.g. divergent thinking, intelligence, giftedness – and many frameworks, mostly derived from cognitive psychology, have tried to explain the mechanisms of the creative process – e.g. blind-variation and selective retention (Campbell, 1960), associative theories (Mednick, 1962), *geneplore* (Finke, Ward, & Smith, 1992), stage models (Wallas, 1926), componential models

(Amabile, 1990), investment models (Sternberg & Lubart, 1991), and contextual models (Csikszentmihalyi, 1996; Gardner, 1993).

In sport, creativity has also seen increased interest from researchers and practitioners (Memmert, 2010), with an exponential growth occurring in the last two decades. The advent of performance analysis and the widespread availability of information on teams and players' behaviours mean that creativity in sport has never been more necessary. It is therefore unsurprising that many stakeholders are trying to develop alternative approaches that increase unpredictability in individual and collective behaviour to be more 'successful' (Yamamoto & Yokohama, 2011 in Torrents et al., 2016).

While important advancements have been made, much remains to be understood about this complex phenomenon. Therefore, the purpose of this review was to critically explore conceptualisations of sporting creativity and methods recommended for its assessment and development, while also suggesting avenues for future research.

Method

Given the emerging nature of the field and the absence of previous reviews on sporting creativity, it was deemed appropriate to conduct a literature review (Grant & Booth, 2009) that focused on a comprehensive search of the existing literature, without assessing the quality of evidence available. Six databases were used: PUBMED, SportDiscus, Web of Science, ERIC, Scopus, and PsycINFO. Search terms were drawn from the seminal literature on creativity and based on Runco's (2014) definition of creativity and its correlates (e.g. intelligence, innovativeness, imagination, inventiveness, originality). With regards to sport, the search used not only the overarching term – sport – but was complemented by Launder and Piltz's (2013) definition of team invasion games, court invasion games and court-divided games. Limited time and human resources led to a focus on these three game categories, where creativity is of 'crucial importance' (Memmert, 2017, p. 479) when compared to individual sports, which tend to fall on the lower end of the complexity continuum (Brown & Gaynor, 1967; Memmert, 2011, 2017). The keyword search was the following: (creativ* OR imaginat* OR intelligen* OR inventive* OR innovative* OR original*) AND (sport* OR football OR soccer OR handball OR volleyball OR ultimate OR hockey OR lacrosse OR tennis OR rugby OR netball OR basketball OR badminton OR futsal OR korfbal). Searches were adapted to the syntax of each database. No participant age limits nor English language limits were applied to searches.

After the initial searches, all titles considered relevant ($n = 196$) were screened to determine their eligibility. One hundred were excluded. The next phase – abstract review – involved screening summaries and comparing them to the inclusion criteria. Eligibility was assessed by using the criteria presented in Table 1. The search did not retrieve any non-English results that would meet all inclusion criteria.

A full-text review of those documents retained ($n = 96$) was conducted, resulting in the exclusion of 65. Additionally, a manual search process was conducted by using both forward and backwards snowballing approaches (Greenhalgh & Peacock, 2005), with two titles ($n = 2$) being included. Finally, the most published authors in the field of creativity in sport (identified via Research Gate and Google Scholar) were contacted to request further information on ongoing research that had passed the stage of data collection

Table 1. Inclusion and exclusion criteria.

Inclusion criteria (studies had to meet all criteria)	(1) Studies referred to sport, in general, or to team invasion games/rugby/court invasion games/court-divided games as defined by Launder and Piltz (2013) (2) Studies were published in peer-reviewed journals and/or conference proceedings and were directly related to creativity or its correlates; (3) Studies were original and published in languages spoken by the authors of this review (English, French, Italian, Portuguese, and Spanish).
Exclusion criteria	(1) Non-peer-reviewed books or book chapters and dissertations were not considered due to lower peer-evaluation standards and difficulties in access; (2) Documents published by institutions with commercial affiliations (e.g. company foundations) were not included;

and on other work that could potentially enrich this review. This resulted in the integration of six more titles ($n = 6$). After the initial search on April 2017, a final search was re-run on August 2018 and seven ($n = 7$) more titles were added to the list. A total of 46 articles (see full description in Table 2) met eligibility criteria (see Figure 1 for a summary of the selection process). Two documents – an academic book and a book chapter – which did not meet the eligibility criteria (peer-review) were added by recommendation of a scholar who considered them key texts, authored by the most prominent researcher in the field. In total, 48 titles were included.

To identify potential patterns in the existing literature, organise them coherently, and reflect on their meaning and implications for research, a narrative structure was adopted and thematic analysis was used (Braun, Clarke, & Weate, 2017). An inductive approach was followed in the coding phase, with the content steering the evolution of the analytical process, which followed Braun et al.'s (2017) proposed six-step process, although not always sequentially.

Higher-order themes (defining creativity, correlates of creativity, assessment of creativity, developing creativity) worked as central organising concepts around which lower-order themes revolve. As part of the active nature of the analytical process, disagreements between authors were resolved through constructive debate that included the opinions of critical friends – a departmental colleague with a background in sport psychology and football, and one of the leading authors contacted during the data collection process – (Berends & Johnston, 2005), until a final structure was agreed (Figure 2).

Results

Defining creativity

This higher order theme presents the review of how creativity is defined. It is comprised of four lower order themes: cognitive traditions, tactical creativity; creativity in sport vs creativity about sport; creativity in sport is emergent.

Cognitive traditions

Initial research on sporting creativity was based on the previous work of cognitive psychologists. Runco (2014) argues that the prevalence of cognitive theories of creativity can be explained by 'an intuitive connection between cognition and creativity and because cognitive research is often very scientific' (p. 1). To date, many different lines of research have

Table 2. Articles meeting the inclusion criteria.

Author(s)	Year	Study Type	Participants	Country participants	Assessment of Creativity
Brown, G.; Gaynor, D.	1967	Position paper	n.a.	n.a.	
Đuriček, M.	1992	Position paper	n.a.	n.a.	
Everhart, B.; Kernodle, M.; Turner, E.; Harshaw, C.; Arnold, D.	1999	Quantitative/Experimental	24	USA	
Memmert, D.	2006	Quantitative/Experimental	2 experiments (Exp.1–33; Exp.2–112)	Germany	Game-test situations
Memmert, D.	2007	Quantitative/Experimental	48	Germany	Game-test situations
Memmert, D.; Furley, P.	2007	Quantitative/Experimental	3 experiments (Exp.1–34; Exp.2–29; Exp.3–16)	Germany	
Memmert, D.; Roth, K.	2007	Quantitative/Experimental	135	Germany	
Memmert, D.; Perl, J.	2009a	Quantitative/Experimental	55	Germany	
Memmert, D.; Perl, J.	2009b	Quantitative/Instrument Validation	42	Germany	
Memmert, D.	2010	Quantitative	195	Germany	
Oh, J. I.; Joung, K.; Kim, H. K.; Choi, H.; Kim, N.; Sung, J.	2010	Qualitative/Survey	52	South Korea	
Greco, P.; Memmert, D.; Morales, J.	2010	Quantitative	22	Brazil	
Lacerda, T.; Mumford, S.	2012	Position paper	n.a.	n.a.	
Memmert, D.; Baker, J.; Bertsch, C.	2010	Quantitative/Survey	72	Germany	
Hopsicker, P.	2011	Position paper	n.a.	n.a.	
Aggerholm, K.; Jespersen, E.; Ronglan, L.T.	2011	Position paper	n.a.	n.a.	
Cavallera, G. M.; Boari, G.; Labbrozzi D.; del Bello, E.	2011	Quantitative/Experimental	61	Italy	TTCT figural series (1989)
Memmert, D.	2011	Quantitative/Experimental	120	Germany	Divergent Thinking Test (Krampen, 1996) – subtest + Video handball- specific divergent thinking
Veraksa, A. N.; Gorovaya, A. E.	2011	Quantitative/Experimental	31	Russia	TTCT verbal and figural
Hristovski, R.; Davids, K.; Araujo, D.; Passos, P.	2011	Position paper	n.a.	n.a.	
Hristovski, R.; Davids, K.; Araujo, D.; Passos, P.	2012	Position paper	n.a.	n.a.	
Memmert, D.; Hüttermann, S.; Orliczek, J.	2013	Quantitative/Experimental	30	Not reported	Video soccer-specific divergent thinking
Bowers, M. T.; Green, B. Ch.; Hemme, F.; Chalip, L.	2014	Quantitative/Survey	99	USA	Abbreviated Torrance Test for Adults (ATTA)
Campos, D.	2014	Position paper	n.a.	n.a.	
Furley, P.; Memmert, D.	2015	Quantitative/Experimental	61	Germany	Video soccer-specific divergent thinking
Memmert, D.	2015	Academic Book	n.a.	n.a.	
Rasmussen, L.; Østergaard, L.	2016	Mixed methods/Experimental	15 players + 1 coach	Denmark	
Igorov, M.; Predoiu, R.; Predoiu, A.; Igorov, A.	2016	Quantitative/Experimental	11	Unreported	Imagination and Creativity Test (Roco, 2004) – first task only

Torrents, C.; Ric, A.; Hristovski, R.; Torres-Ronda, L.; Vicente, E.; Sampaio, J.	2016	Quantitative/Experimental	44	Unreported	Small-sided games
Arslan, K. S.; Akpunar, F.; Ulucan, K.	2016	Position paper	n.a.	n.a.	
Harrison, C.	2016	Qualitative/Auto-ethnography	1	Australia	
Moraru, A.; Memmert, D.; van der Kamp, J.	2016	Quantitative/Experimental	2 experiments (Exp.1–57; Exp.2–56)	Unreported	
Santos, S. D. L.; Memmert, D.; Sampaio, J.; Leite, N.	2016	Position paper	n.a.	n.a.	
Martin, J.; Cox, D.	2016	Qualitative/Biographical	Multiple	Canada/USA	
Tanggaard, L.; Laursen, D.; Szulevicz, T.	2016	Qualitative/Material Biography	n.a.	n.a.	
Santos, S.; Jiménez, S.; Sampaio, J.; Leite, N.; Padulo, J.	2017	Quantitative/Experimental	40	Portugal	TTCT – figural version + small-sided games (GK + 3v3 + GK)
Ilundain-Agurruza, J.	2017	Position paper	n.a.	n.a.	
Rasmussen, L.; Østergaard, L.; Glaveanu, V.	2019	Position paper	n.a.	n.a.	
Leso, G.; Dias, G.; Ferreira, J. P.; Gama, J.; Couceiro, M. S.	2017	Quantitative/Survey	34 coaches + 118 players	Portugal	
Orth, D., van der Kamp, J.; Memmert, D.; Savelsbergh, G.	2017	Position paper	n.a.	n.a.	
Memmert, D.	2017	Book chapter	n.a.	n.a.	
Richard, V.; Abdulla, A.M.; Runco, M.	2017	Quantitative/Experimental	208 athletes (94 males, 114 females)	Canada	Runco Creative Assessment Battery
Santos, S.; Coutinho, D.; Goncalves, B.; Schollhorn, W.; Sampaio, J.; Leite, N.	2018	Quantitative/Experimental	40	Portugal	Small-sided Games (GK + 5v5 + GK)
Hendry, D.; Williams, A.M.; Hodges, N.	2018	Quantitative/Survey	102	UK	
Furley, P.; Memmert, D.	2018	Quantitative/Lab-based Experiment	120 (39 women) players	Germany	
Kempe, M.; Memmert, D.	2018	Quantitative/Match-analysis	153 games, 311 goals	Worldwide	
Roca, A.; Ford, P.; Memmert, D.	2018	Quantitative/Experimental	44 male players	England	Life-size soccer-specific divergent thinking
Hüttermann, S.; Nerb, J.; Memmert, D.	2018	Quantitative/Experimental	30 male players	Germany	Video-based divergent thinking task

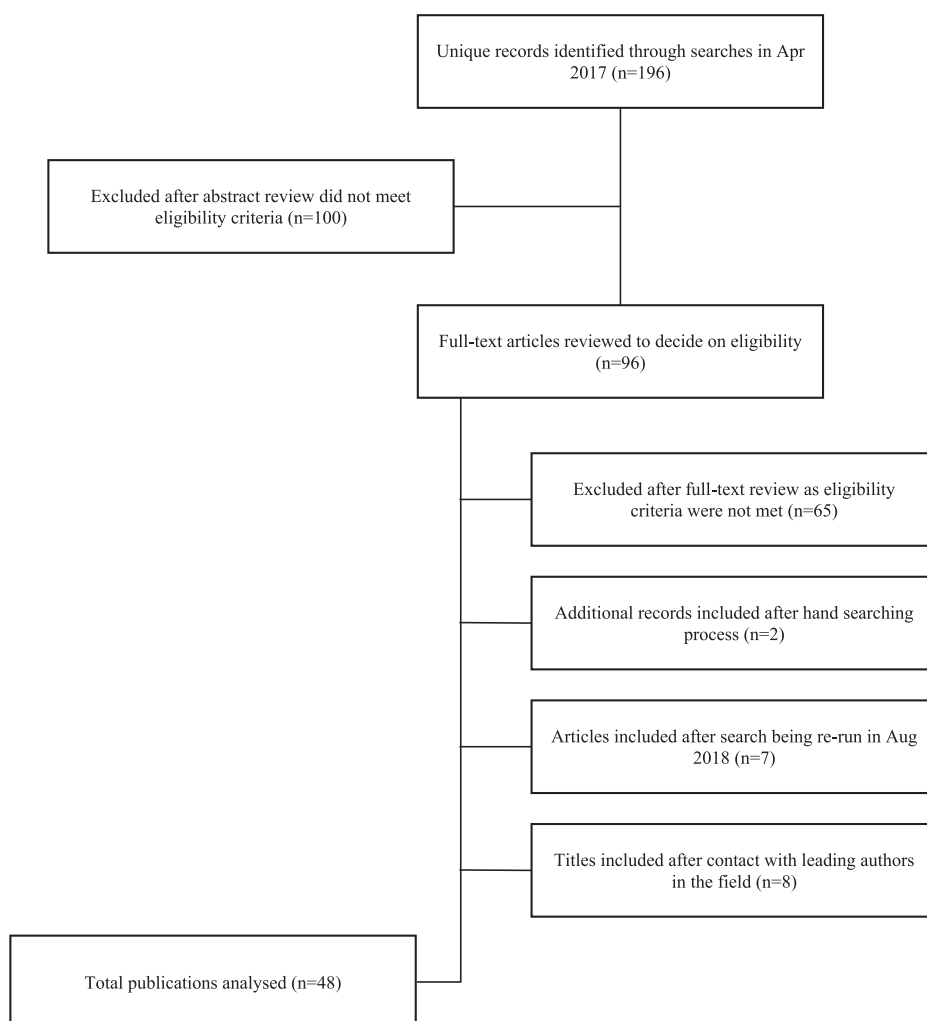


Figure 1. Selection process.

been explored – intelligence, giftedness, divergent thinking (the ability to generate different possible solutions to a problem), and so forth. These research streams have also inspired work on sporting creativity (e.g. Hopsicker, 2011; Igorov et al., 2016; Memmert, 2006; Memmert & Roth, 2007).

A plethora of definitions of creativity has been proposed. Sternberg and Lubart (1999) suggested that for an action to be creative it must be novel and useful. Boden (2004) and Simonton (2012) added a third criterion: surprise. With regards to the assessment of the creativity of an individual, (Guilford, 1967) introduced three foundational dimensions: fluency (the ability to generate several responses), flexibility (the ability to generate different categories of responses) and originality (the ability to generate unusual responses). Memmert, the most cited author in the field, who participated in 44% of publications included in this review, frequently uses Guilford's (1967) dimensions to assess creative solutions in sport.

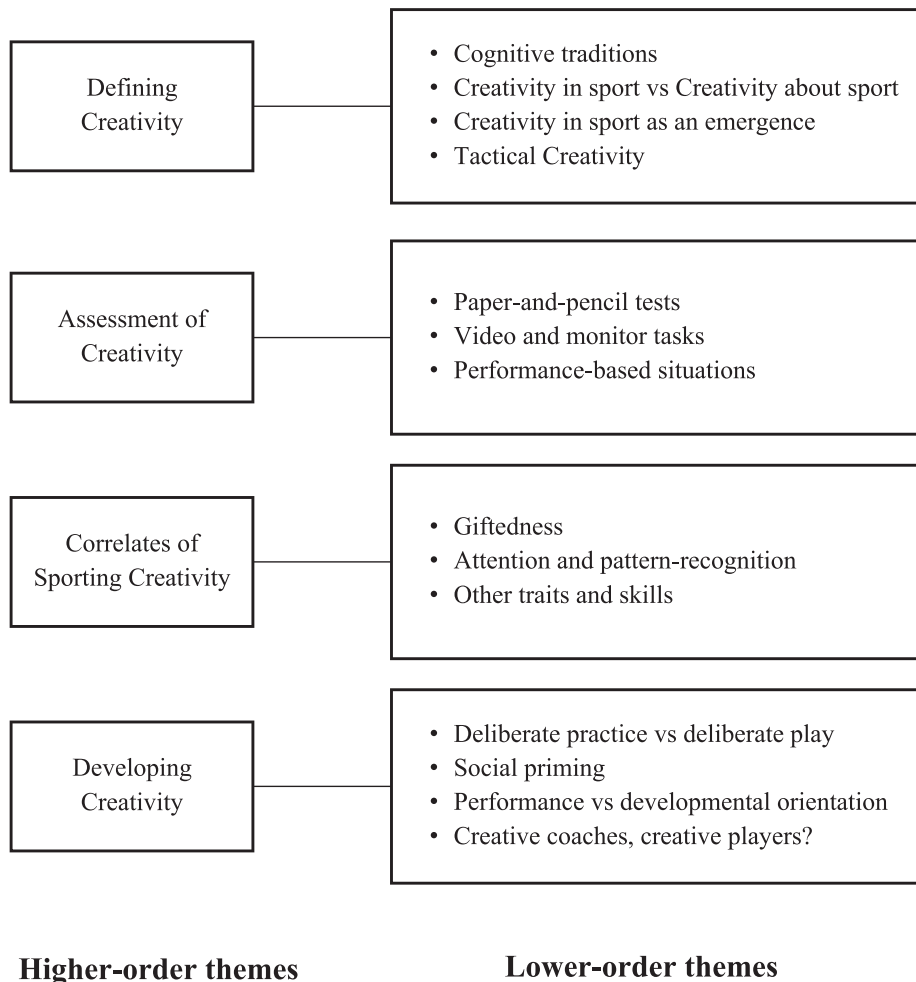


Figure 2. Distribution of higher and lower-order themes.

Tactical creativity

Memmert adapted Sternberg and Lubart's investment model (1991) to coin a definition of *tactical creativity*, which refers to 'those varying, rare, flexible decisions that play an important role in team ball sports like football, basketball, field hockey and handball' (Memmert, 2011, p. 94). Tactical creativity (or divergent tactical thinking) differs from game intelligence or convergent tactical thinking, that relate exclusively to the selection of the most effective solutions for a given problem (Memmert, 2010). Rather, tactical creativity emphasises the ability of players to generate the highest possible number of different motor solutions for a problem. Furthermore, it is proposed that tactical creativity can only occur in the offensive phase of a game, and not in defence (Kempe & Memmert, 2018; Memmert, Baker, & Bertsch, 2010). The focus on attacking players and play has influenced the research conducted. For example, as part of participant selection, Memmert et al. (2010) asked coaches to identify the most creative attackers and least

creative defenders. In another study, Kempe and Memmert (2018) focused on the creativity of the last eight actions leading to a goal scored in open play in football World Cups and European Championship. Based on their findings they concluded that creativity is particularly important for attackers and that creativity is 'a decisive factor for success in soccer' (2018, p. 4).

In contrast, professional football coach Jose Tavares, interviewed by Tamarit (2016) contends that all players can be creative, in any phase of the game, with and without the ball. That creativeness should be shaped and evaluated against the specific requirements of each player's position. For example, defenders can produce creative actions within the specificity of their role, while a winger can do the same, perhaps through different strategies, more adapted to the position's requirements and dependent on the team's overarching game model – a way of playing. The interdependence between attackers and defenders is supported by the work of Aggerholm, Jespersen, and Ronglan (2011), who performed a contextual analysis of the feint in association football. They concluded that other than self-awareness and the cultivation of embodied habits, to be creative 'it is also necessary to be absorbed in the other and transcend his or her expectations' (2011, p. 343). Consequently, the emergence of tactical creativity may depend not only on individual or cooperative efforts but also from this relationship with the opposition – the *duel*.

Creativity in sport vs Creativity about sport

An aspect of creativity conceptualisation that lacks consensus is the role of performance. Some suggest creativity depends on the final product, creative performance (e.g. Kaufman & Sternberg, 2007), others emphasise the ability to generate ideas, even if these are not materially expressed (see Runco, 2014 for a comprehensive insight on this discussion). Brown and Gaynor (1967) proposed that sporting creativity needs to be expressed through non-verbal motor skills (creativity *in* action as opposed to creativity *about* action). These creative motor skills can be expressed individually or collectively, and the level of creative potential of an action depends on its complexity. For example, running 100 m in a straight line has less creative potential (i.e. less different possibilities for action) than playing a game of basketball.

Brown and Gaynor (1967) also argued that the creative processes in sport operate much in the same way – preconscious incubation preceding the emergence of the creative action – as those of other areas which do not require physical exertion (e.g. writing, composing). Recent neuroscientific findings on the impact of mechanisms of brain inhibition on creative performance suggest this may not be the case. In their reticular-activating hypofrontality (RAH) model of acute exercise, Dietrich and Audiffren (2011) argue that the brain uses two different cognitive systems to acquire and represent information: implicit and explicit. The explicit system deals with abstraction and complex problem-solving, being linked to more traditional forms of creative expression – e.g. writing, composing. It is rule-based, relates to conscious awareness, and can be expressed verbally.

On the other hand, the implicit system, to which motor skills are related, relies heavily on procedural knowledge, which cannot be verbalised, depending therefore on task performance to be expressed. Therefore, sporting creativity, in part, may operate differently to creativity in some other domains because the expression of creativity is through action rather than about action. Furthermore, the unstructured nature of many sports, particularly

team sports, demands constant reaction and adaptation to different stimuli. Real-time creativity is limited by time constraints and is necessarily spontaneous (Harrison, 2016). This has implications for the conceptualisation and assessment of sporting creativity as well as the design of interventions to facilitate creativity in sport.

Sporting creativity as an emergence

The idea of body-mind integration (Brown & Gaynor, 1967; Campos, 2014; Hristovski, Davids, Araújo, & Passos, 2011; Hristovski, Davids, Passos, & Araújo, 2012; Ilundáin-Agurruza, 2017; Krein & Ilundáin-Agurruza, 2017), as opposed to the traditional privileging of the mind, has important implications for the conceptualisation and development of sporting creativity. Challenging established ideas of the brain as the trigger of all action, Gibson (1979) argued that information, i.e. spatiotemporally patterned energy flow from the environment, is the key element to locomotion and manipulation and that the interaction between the individual and the environment was critical. In this perception-action system, meaning comes from the individual's ability to detect information in the environment (Araújo, Hristovski, Seifert, Carvalho, & Davids, 2017). Furthermore, physical exploration of the landscape of action possibilities may result in the discovery or emergence of a novel action.

Orth, van der Kamp, Memmert, and Savelsbergh (2017) also emphasise the importance of adaptability to the environment in motor creativity, which they define as 'new ways of acting adaptive or acting adaptively in new situations' (p. 2). As such 'the emergence of highly novel movement forms requires a self-organising system which, under suitable boundary conditions, can create new behavioural structures' (Hristovski et al., 2011, p. 177). These constraints offer the individual opportunities for action, which Gibson (1979) termed *affordances*. For example, Tanggaard, Laursen, and Szulevicz (2016) showed that changes in equipment (material constraints), in this case, the material that handballs were made from (synthetic polyurethane compared with leather), led to new possibilities for creative expression.

Correlates of sporting creativity

Following the tradition of cognitive psychology, earlier investigations of sporting creativity attempted to identify isolated variables that contribute to increased creativity. Researchers have examined a relatively limited range of variables, which are discussed under three lower-order themes: giftedness, attention and pattern recognition, and other traits and skills.

Giftedness

The research examining giftedness and creativity has demonstrated that gifted children (IQ > 130) tend to express creative behaviour earlier than their non-gifted peers. For example, Memmert (2006) investigated the creative performance of children who, once a week and for six months, underwent a sports enrichment programme which consisted mainly of diversified team ball sports practice (using feet, hands, and a hockey stick) in game forms. Memmert found that while there was no significant improvement in the gifted control group, the gifted experimental group showed a significant increase in creative performance after six months. The non-gifted experimental group did not show a

significant improvement as a result of the intervention. However, in a different study with non-gifted children, Memmert and Roth (2007) showed a 40% improvement in creative performance after a 15-month training period. Memmert (2006) explained the accelerated improvement of the gifted group in the shorter programme was a result of 'faster automation of individual thought processes ... This frees attention capacity for other tasks' (p. 108). Therefore, creativity is not a characteristic of only gifted children, it can be developed by others but may take longer.

Attention and pattern-recognition

Brown and Gaynor (1967) highlighted the crucial role of extreme awareness in creativity by suggesting that 'the athlete who is most creative is most aware, most in tune with reality as it exists. Being aware of the single large problem (the game), he [*sic*] is able to recognise and to act on smaller problems which arise continually' (1967, p. 157). It is, therefore, perhaps, unsurprising that breadth of attention as a correlate of creativity has received the most research attention (e.g. Furley, Memmert, & Heller, 2010; Memmert, 2007; Memmert & Furley, 2007; Moraru, Memmert, & van der Kamp, 2016). This work has focused particularly on inattention blindness, which relates to the diversion of attention where people fail to notice something unexpected, even when it is in their field of view (Memmert, 2007).

In a series of experiments, Memmert and Furley (2007) examined inattention blindness in youth handball players, using a video task. They were interested in participants' ability to notice an unmarked player that appeared unexpectedly in the game and the effect of different instructions and actions of the unmarked player. They found that when there were no other conditions, 45% of participants failed to notice the unmarked player, however, when one group was given closed-end instructions 83% of participants failed to notice the unmarked player. This contrasted with only 17% of those participants who were not given these instructions failed to notice the player. Furthermore, when the unmarked player waved his arms only 6% of participants failed to notice him. Connecting inattention blindness with creative performance, Memmert (2011) examined the relationship between attention and experience in both general and sporting creative performances. The study involved skilled (with a previous degree of experience in team invasion sports) and non-skilled (with no previous experience) handball players aged between 7 and 13 years. He found that inattention blindness was higher in the youngest children (7 years of age) and performance of attention tasks improved in children between the ages of 8 and 13 years. Memmert also noted an evident plateau in the children between 10 and 13 years, which was attributed to the decrease in the 'absolute number and density of synapses as one grows older, making it harder to improve creative thinking' (Memmert, 2011, p. 93).

Adding further evidence of the relationship between attention and creative performance, Moraru et al. (2016) manipulated participants' breadth of attention. Participants in the broad focus group were more inclined to use more different modes of locomotion (flexibility), but not invest as much time on finding solutions within a particular mode (persistence). A broader focus did not significantly enhance originality, which is in contrast to results of previous studies on divergent thinking (e.g. Memmert, 2011). This can be explained by the increased difficulty of performing a wider range of motor skills (which is largely limited by existing motor ability) in comparison to thinking (ideation) skills: 'if

motor ability is insufficient (e.g. only a few people can walk on hands), then the thought of action cannot be performed' (p. 10).

Furthermore, Memmert (2007) demonstrated that a six-month attention-broadening training programme had a positive effect on the creative performance of children. He compared an attention-narrowing group (with teachers giving explicit tactical instructions and corrections constantly during play) with an attention-broadening group (with teachers only giving general advice about the games and their rules, and not providing any kind of feedback during play). Memmert found that only the attention-broadening group considerably improved their general creative performance.

An association between creative performance and visual search behaviours has also been demonstrated by Roca, Ford, and Memmert (2018) who used a portable eye-movement registration system to examine creativity in decision-making and visual search behaviours of expert football players during simulated 11-a-side matches. They found that more creative players, when compared to their less creative counterparts, displayed a broader attentional focus which included a higher number of fixations, but of shorter duration. They were also able to perceive earlier the location of unmarked teammates and opponents.

Other traits and skills

The relationships between creativity and a small number of other traits and skills have been examined. These include: working memory, morning-eveningness personality, coping, and regulatory focus. Researchers have examined the role of working memory in sporting creativity – both creative thinking and creative action, however, no evidence has been found that working memory interferes with creative ability (Furley & Memmert, 2015; Moraru et al., 2016). With regards to morningness-eveningness personality, Cavallera, Boari, Labbrozzi, and Del Bello (2011) found that participants with an intermediate (not morningness nor eveningness-oriented) personality type had significant positive correlations between the number of hours of sport activity per week and scores of elaboration (measured through the Torrance Test of Creative Thinking – TTCT). Creative thinking performance, however, was independent of gender and personality typology. In their study of junior female handball players, Igorov, Predoiu, Predoiu, and Igorov (2016) found a significant positive correlation between fluency and positive reinterpretation as a coping strategy but the relationship between coping and flexibility was not significant. They speculated that these findings relate to situations in which athletes try to find positive aspects in undesirable situations, often through the recollection of past successful performances. In relation to regulatory focus, Memmert, Hüttermann, and Orliczek (2013) found that adult male football players with a promotion (aspirational) focus performed better in a sport-specific divergent thinking video task than those with a prevention (duty-oriented) focus, which corroborates repeated claims (e.g. Āuriček, 1992; Hopsicker, 2011) that risk-taking and open-mindedness enable creative behaviour and an avoidance focus may constrain creative behaviour. Hüttermann, Nerb, and Memmert (2018) have recently replicated the earlier study by Memmert et al. (2013), to investigate the relationship between regulatory focus, expectations and performance, among a more experienced sample. While promotion focused players displayed, once again, significantly higher values in terms of creativity, there was no main effect on expectation nor any significant interaction.

Assessing creativity

This higher order theme captures the methods employed to assess creativity in sporting environments. The theme comprised three lower order themes: paper-and-pencil tests, computerised and video tasks, and performance-based situations.

Paper-and-pencil tests

Several different pencil and paper tests have been used to assess sporting creativity, either in part or in full. These tests are largely adapted from psychological tests and assess general creative thinking. One of the most commonly employed measures of creative behaviour is the TTCT. It exists in two formats – figural and verbal – and assesses creative thinking through four components fluency, flexibility, and originality, plus elaboration (amount of detail in responses) (Bowers, Green, Hemme, & Chalip, 2014; Cavallera et al., 2011; Santos, Jimeénez, Sampaio, Leite, & Padulo, 2017; Veraksa & Gorovaya, 2011). The main advantage of the TTCT is that it is one of the few valid and reliable tests of divergent thinking (Kim, 2011). Others tests that have been employed include Roco's (2004) Imagination and Creativity Test (e.g. Igorov et al., 2016). and Krampen, Freiling, and Willems's (1996) Divergent Thinking Test (M Emmert, 2007). However, the very small sample ($n = 11$) and lack of detailed information on Roco's (2004) test suggest limited validity and reliability of Igorov et al.'s (2016) results. Moreover, all tests enumerated are tests of generalist *thinking* expressed verbally or through drawing, not a measurement of physical *doing*.

Video and monitor tasks

M Emmert and colleagues (Furley & Memmert, 2015; Hüttermann et al., 2018; Memmert, 2011; Memmert et al., 2013; Roca et al., 2018) are the only researchers, to date, who have used video and monitor tasks to examine the relationship between sporting creativity and other cognitive skills or traits – e.g. attention, working memory, visual search behaviour, regulatory focus. The tasks involve participants watching videos of sporting gameplay and then being asked to provide possible attacking options. For example, in a handball-specific task (M Emmert, 2011), participants watched five videos of a handball game involving four attackers and four defenders. After one minute, the video would stop, and the last frame would remain on the screen. Participants would then be asked to imagine they were an attacker and indicated all potential opportunities that could lead to a goal. The proposed options are assessed for creativity using traditional criteria of originality, flexibility and fluency. A football-specific video task has also been developed. It is composed of 20 different football attacking scenes from 46 Bundesliga 1 and 2 matches (Germany, season 2010/2011), selected by experienced certified coaches (Furley & Memmert, 2015; Hüttermann et al., 2018; Memmert et al., 2013).

Video tasks are more representative of sport when compared to paper-and-pencil tests. Memmert (2015) suggests that although standardised video tasks are less complex, they have less confounding variables and the selection of clips shown to participants can be adjusted to reflect specific situations. Roca et al. (2018) acknowledged the limited physical realism of these tasks, which 'might alter the natural role of the underlying perceptual-cognitive processes underpinning players' creative behaviour' (p. 2), proposing instead the

adaptation of Furley and Memmert's (2015) task to life-size-video based simulations in which participants had to play an actual ball in addition to providing a verbal response.

However, despite in different degrees, video tasks still focus on divergent thinking as the only measure of creative ability. Convergent thinking, i.e. the orientation 'toward deriving the single best (or correct) answer to a clearly defined question' (Cropley, 2006, p. 391) also contributes to creative insights (Dietrich & Haider, 2017) but is not considered. Furthermore, like paper-and-pencil tests, most of these tasks do not allow for a realistic assessment of sporting creativity *in action*.

Performance-based situations

Performance-based situation tests, where participants' actual performance is assessed for creativity, have been developed and employed by researchers (e.g. Memmert, 2007; Torrents et al., 2016). Moraru et al. (2016), used an agility ladder in a divergent doing task. Participants were asked to perform the highest possible number of different actions on the agility ladder, i.e. using their feet and hands for stepping, hopping, skipping, walking, and so forth. While this approach is arguably more representative of creativity in action compared with paper and pencil tests and video tasks when considering its use, researchers perhaps should ask how well it represents creativity in specific sporting contexts.

The use of small-sided formats is an alternative that has enhanced representativeness in comparison to the agility ladder, as it tests players in actual game forms. Criteria of originality, flexibility, and fluency are used to assess performance, with scores being averaged into a single measure of creativity. For example, Memmert and Roth (2003) created *game-test situations (GTS)* where creative performance is assessed through *orienting and supporting* and *identifying gaps* actions of participants during small-sided games (for a detailed description see Memmert, 2006). In Memmert and Roth (2007), children performed with hands, feet and a hockey stick, but in other studies, only one of the skills was evaluated (e.g. Memmert, 2010).

Along similar lines, Torrents and colleagues (2016) examined differences in exploratory behaviour motivated by numerical superiority, equality, or inferiority with 44 football players (22 professional and 22 amateur players) using small-sided games (4 vs 3, 4 vs 5, and 4 vs 7). An observation instrument (adapted from Costa, Garganta, Greco, Mesquita, & Maia, 2011; Owen, Twist, & Ford, 2004) was used to record the possible actions from attackers with the ball (e.g. run to the ball, control, pass, shoot) or without the ball (e.g. wall, support, unmark) and from defenders (e.g. press, delay, dissuade). Santos and colleagues (Santos et al., 2017; Santos et al., 2018) also used small-sided games and an observation instrument (Creativity Behavior Assessment in Team Sports – CBATS) to assess in-game individual (passing, dribbling and shooting actions) and collective behaviour (regularity of team movements and distance between players which was assessed through GPS measurements). A creative behaviour score was established and included attempts – defined as the effort to perform different actions, successful or unsuccessfully –, fluency – the ability to execute the highest possible number of successful movement actions, and versatility – the ability to generate a diversity of actions within the same category, e.g. different types of passing or shooting.

The design and use of game-based situations and accompanying observation tools to assess sporting creative behaviour is an important development with regards to task representativeness and ecological validity, particularly when it includes assessment of

individual as well as collective behaviours (e.g. Santos et al., 2017, 2018). Only Torrents and colleagues (2016) included creative defensive behaviour, although they did not include goalkeepers, which again limits our understanding of sporting creativity in all phases of the game. Furthermore, Santos and colleagues (2017, 2018) and Memmert and Roth (2007) measured creativity in situations of numerical equality, or superiority, however, Torrents and colleagues (2016) found that numerical inferiority might lead to greater exploratory behaviour.

Memmert (2015, 2017) has proposed the use of game observation of real matches as a new standard to evaluate tactical performance due to its 'very high ecological validity' (p. 482). These observations can be aided by game play protocols which combine quantitative (e.g. frequency of certain behaviours) and qualitative components (the subjective yet knowledgeable opinions of experts). Finally, Memmert (2017) suggests that technology can play an important role in analysing creative behaviour. For example, using neural networks to categorise action processes in team sports (e.g. Memmert & Perl, 2009b).

Developing creativity

Understanding how to develop creative players is one of the key ambitions of academics and practitioners. It is then unsurprising that a growing body of literature on the topic is emerging. Four lower-order themes were developed through the review: deliberate practice and deliberate play; social priming; programmes for the development of sporting creativity; the central role of coaches in creative development.

Deliberate practice, deliberate play

In line with research conducted on expertise, skill acquisition and talent development (Davids, Güllich, Shuttleworth, & Araújo, 2017), six studies on sporting creativity (Bowers et al., 2014; Greco, Memmert, & Morales, 2010; Hendry, Williams, & Hodges, 2018; Martin & Cox, 2016; Memmert, 2006; Memmert et al., 2010) have devoted attention to deliberate practice and deliberate play. Deliberate practice is 'the engagement in practice activities with a clear goal of improving a specific aspect of performance beyond its current level' (Ericsson, 2017, p. 4). In turn, deliberate play, which is usually fostered during sampling years (ages 6–13), does not intentionally focus on performance improvement, prioritising instead 'developmental physical activities that are intrinsically motivating, provide immediate gratification, and are specifically designed to maximize enjoyment' (Berry, Abernethy, & Côté, 2008, p. 687).

After a six-month intervention, Memmert (2006) found that deliberate play had a positive impact on the tactical creative performance of gifted children. Similarly, in a field study involving Brazilian youth basketball players, Greco et al. (2010) discovered that unstructured play significantly improved measures of tactical creativity and tactical intelligence (i.e. finding the most appropriate solution for a problem). In contrast, in an examination of professional youth football academy coaches and players' skill-ratings over a period of 5 years, Hendry and colleagues (2018) found that while ratings of top players were positively related to practice, they were negatively related to the proportion of play vs practice. Hours spent in play were not correlated with ratings of any skill, including creativity. The authors concluded (2018, p. 7) that 'there may be benefits

to participation in coach-led practice and play from an early age, potentially due to the need to accumulate a high volume of sport-specific activity, as well as sufficient variations in practice’.

Despite some studies highlighting a more pronounced influence of deliberate practice or deliberate play on creative development, most authors (e.g. Bowers et al., 2014; Memmert et al., 2010) concur that the combination of both strategies may be essential in the development of sporting creativity. This is further supported by work developed by Richard, Abdulla, and Runco (2017), who explored the influence of skill level, experience, and hours of sport training and participation on everyday creativity (e.g. divergent thinking related to diary and distraction management, creative attitude and values) on a sample of 208 Canadian athletes (21 intermediate, 73 advanced, and 114 experts, including Olympic and world-class competitors) aged between 14 and 37, across 17 different sports. Richard and colleagues found that expert athletes displayed a significantly higher cognitive flexibility, while athletes who engaged in a higher number of sports were significantly more creative in comparison to those who only practised one sport.

Finally, having explored the early life experiences of former NBA star Steve Nash, Martin and Cox (2016) found other factors that may have contributed to Nash’s creative development: parental influence, intrinsic motivation, peer support, and self-determination. Bowers and colleagues (2014) suggest that improving creativity does not require ‘a complete reimagining of entrenched youth sport development models’ (p. 325) but could be achieved through the redistribution of time allocated to each activity.

Social priming

To date, only one study (Furley & Memmert, 2018) has examined the impact of social priming, i.e. the use of world-class creative players as role models, on creative thinking. This study, with amateur adult football players, demonstrated that asking participants to write down the characteristics of the creative player (e.g. Lionel Messi) and imagine a typical situation that this player would be involved in led to more creative responses to attacking scenarios. Furley and Memmert concluded that it is possible to prime creative thinking in football players, by activating ‘cognitive representations of creativity which in turn can activate associated mindsets, information processing modes, and response tendencies’ (2018, p. 7).

Programmes for the development of creativity

While much of the research on sporting creativity has focused on isolating traits and processes, there have been recent efforts to provide macro-structures (e.g. frameworks, programmes) for the development of creativity. Three such programmes are the Tactical Creativity Approach (Memmert, 2015), the Creative Development Framework (CDF), which includes the Skills4Genius programme (Santos et al., 2016) and The Creative Soccer Platform (TCSP) (Rasmussen & Østergaard, 2016). Preliminary research, although limited, suggests these programmes do develop sporting creativity.

The Tactical Creativity Approach (TCA) is the result of Memmert’s extensive research on sporting creativity and represents the translation of his key findings into an operational framework. The TCA (Memmert, 2015) is composed by 6 D’s: deliberate-play, 1-dimension games, diversification, deliberate coaching, deliberate motivation, and deliberate practice. Memmert (2015, 2017) proposes that special emphasis is placed on the first four D’s during

earlier stages of player development. Deliberate play and deliberate coaching relate to unstructured play without coaches or teachers actively providing instructions to players, in order to allow the latter to come up with multiple different solutions while keeping a wide attentional focus. 1-dimension games are basic game forms specifically aimed at improving tactical components. They are based on 'clearly defined games ideas, fixed number of players, and defined rules and environmental conditions' (Memmert, 2015, p. 51). Diversification refers to the contact with different sports and different stimuli within the same sport (e.g. playing with balls of different sizes, shapes, and materials in football). At more advanced stages of player development, Memmert (2015) highlights the importance of deliberate motivation and deliberate practice. With regards to the former, the TCA favours promotion instructions, which according to earlier research by Memmert et al. (2013), may favour creative expression. Finally, the latter can be developed through sport-specific, task-centred practice 'to explore seldom but adequate solutions' (p. 96).

The Creative Development Framework (CDF) is another model for the long-term development of creative behaviour in team sports. Development is divided into five stages where free-play and diversification are encouraged at the earlier stages of youth development, advocating a transition to specialisation that is completed around the age of 16. The CDF puts an emphasis on fundamental movement skills (Smith, 2014), fundamental game skills (Smith, 2014), non-linear pedagogy (Chow, 2013), differential learning (Schöllhorn, Mayer-Kress, Newell, & Michelbrink, 2009), teaching games for understanding (Tan, Chow, & Davids, 2012), and constraints-led approach (Hristovski et al., 2011), as ways of developing creative behaviour in sport. One of the key aspects of this model is the belief that sporting creativity does not depend solely on skill mastery, but also relies on the ability to think creatively.

The CDF has been partially tested ('Explorer' phase only – Skills4Genius programme) with Portuguese primary school children (Santos et al., 2017). Findings suggest the programme leads to improvements in general creative thinking, increased fluency, elaboration, and originality. Effects on motor skills are less clear. However, improvements were demonstrated for in-game creativity (attempts, fluency, and versatility). Another empirical study based on the CDF (Santos et al., 2018) examined the impact of differential learning, with an emphasis on small-sided games, as an enhancement strategy for creative behaviour in youth football. While the control groups did not alter their practice routine, the experimental groups took part in a differential learning programme, with three 30-minute training sessions per week, taking place at the beginning of their team's training session. The training programme involved playing small sided games with a constant variation of conditions such as balanced and unbalanced number of players, different balls, pitches with different shapes, and numerous body constraints (e.g. visual occlusion, hands behind head). Creative performance was assessed through the CBATS (Santos & colleagues, 2017). The experimental group demonstrated a significant reduction in failed actions and increased attempts and versatility.

The Creative Soccer Platform (TCSP) (Rasmussen & Østergaard, 2016), based on Byrge and Hansen's work (2009, 2014) in educational settings, has four pillars: task-focus, parallel thinking, lateral thinking, and no experienced judgement. The programme focuses on 'establishing a creative environment (i.e. a playful atmosphere) by facilitating creative processes (i.e. soccer-specific creativity exercises) where players try to develop creative

products (e.g. new feints, dribbles or first touches) and train their creative abilities (not fearing to make mistakes)' (p. 9). The impact of the TCSP was assessed through a focus group with some players and an interview with the coach. Rasmussen and Østergaard (2016) identified some limitations such as initial resistance to change and the difficulty of operating in a hybrid system (after the 'creativity' training in the first half of the session, players resumed normal structured training). They found that experiencing a variety of actions with the ball during the creativity exercises increased the chances of players trying different actions in competitive matches.

Creative coaches, creative players?

To date, only two studies have examined coaches' perceptions of creativity and its development. Distributing open-ended questionnaires to Korean football coaches completing their C and B licenses, Oh et al. (2010) found that coaches associated 'unpredictability, adaptability, improvisation, and mediating' (2010, p. 65) with football creativity and prioritised the promotion of fundamental skills and self-determination as tools for its development.

Moreover, coaches indicated a lack of knowledge on how to teach creativity and revealed that they used personal experiences to overcome that gap. This study also identified several barriers to the improvement of sporting creativity, such as autocratic coaching styles, a focus on results which put coaches under pressure to win matches, the league systems, and a lack of appropriate training facilities. Leso et al. (2017) examined football coaches' perceptions of creativity and game intelligence through a questionnaire containing a set of closed questions. They found that coaches associated creativity with magical thinking.

Discussion

The purpose of this paper was to critically examine the existing literature on sporting creativity, exploring conceptualisations of sporting creativity and methods recommended for its assessment and development. While the first paper reviewed dates from 1967, the last two decades have seen an exponential increase in the number of publications on sporting creativity. This review contributes to our understanding of sporting creativity by providing the first review of research on defining, assessing and developing creativity in sport.

A single definition of sporting creativity has yet to be universally accepted. Understanding what is meant by creativity is considered 'the single most fundamental problem in the field' (Simonton, 2012, p. 97). While there is an overall convergence on the general criteria that make an action creative (i.e. novel and useful), deciding on the appropriateness and novelty of an action or idea is invariably conditioned by both the context and the experiences and beliefs of those judging them. With regards to the context, what is valued in a given time and location will determine whether an action is indeed novel, surprising, and appropriate for a given situation. For example, even the most common actions performed by a handball player in Germany, where the sport is widely developed, are likely to be considered original by most of the British population, who are almost entirely unfamiliar to the sport. The degree of appropriateness or originality of an action depends on who is judging it and on his/her previous experiences. For example, a football coach that favours

positional attacking may consider the Spanish style of playing *tiki-taka* a useful strategy, while another, favouring a direct game, may disagree.

The use of experts has been proposed and widely adopted to mitigate the relativism of evaluations, – including in sporting creativity (e.g. Hendry et al., 2018; Memmert et al., 2013; Torrents et al., 2016). This solution is criticised by Runco and Chand (1994), who question why expert ratings should deserve higher credit than self-reported, peer or teacher evaluations. However, it must be recognised that coaches and scouts are central figures in the identification and development of creativity in sport, having key roles in shaping players' experiences and opportunities. Future research focused on the evaluation of creative behaviour in sport should consider extending the use of expert ratings beyond the criteria of originality (current practice), to include the assessment of adequateness of solutions as well.

Conceptualisations of creativity must also be clear about the social-cultural context in which the action takes place and provide detailed justifications on the choice of judges, as these elements influence what is deemed creative. Stein's (1953 in Runco & Jaeger, 2012) definition of creative work – 'a novel work that is accepted as tenable or useful or satisfying by a group in some point in time' (p. 94) could be used to upgrade the existing definition of tactical creativity coined by Memmert (2011). This definition should also consider both attacking and defensive play, and the interdependence between attackers and defenders.

A constructive alignment between definitions of creativity and research methods used to assess creativity should be considered. The use of game-based situations has improved ecological validity, albeit in quasi-naturalistic settings (researcher-controlled, non-competitive). Definitions of creativity operationalised through these assessments should also be reconsidered (e.g. creativity only occurs in attacking play), together with potential limitations to creative expression 'imposed' on participants through task design, e.g. inferiority vs superiority of numbers in games.

Because of the clear influence of cognitive traditions in much of the research on creativity in sport, creativity definitions and evaluations have privileged the thought process over the ability to act, limiting the understanding of *doing* (performance) as an integral feature of sporting creativity. Performance, real-time expression, and a reliance on the brain's implicit system suggest that creativity should be conceptualised as 'in sport' – in action – rather than 'about sport' and assessed and developed accordingly. If a driver for examining sporting creativity is the desire to increase unpredictability in performance and therefore a competitive advantage, then the final product of the creative process – performance of the creative action – ought to be a critical feature of how creativity is conceptualised.

With regards to developing sporting creativity, research is advancing towards more integrative approaches that can be implemented over an extended period and accompany players' developmental journeys. However, despite encouraging initial results, more research is needed to evaluate the long-term effects of these programmes on creativity due to creativity developing over a long period (Memmert & Roth, 2007). Furthermore, while Memmert's TCA (2015) proposes a holistic framework, creativity training in both the CDF and TCSP was limited to a small part of training sessions. Future research could explore the effects of programmes in settings that do not treat creative training as an appendix or an isolated section of a session but instead adopt the philosophical

underpinnings of creative development as an orienting matrix for the whole session planning, delivery and reflection.

A much more detailed description of the tasks executed by players during training programmes is needed, to allow for more accurate categorisation of what constitutes deliberate practice and deliberate play. The way both concepts are depicted in existing research is too broad. Also, participant sampling could be more consistent to increase the validity of findings. For example, in Memmert et al.'s (2010) study direct comparisons were made between players with different roles – attackers and defenders – who occupy different areas of the pitch and perform different actions, which may have had an impact on results. It must also be recognised that while important, strategies like deliberate practice or deliberate play are merely partial influences in the overall development of children. For example, in Bowers and colleagues' (2014) work, sporting activities accounted for only 30% of the total leisure time of participants. Consequently, interdisciplinary and multi-dimensional approaches that look beyond the sporting arena are recommended, to establish a more complete description of the development of creativity across the lifespan.

The existing literature indicates that sporting creativity can be trained. It does not, however, support a single strategy for its development. So far, a balance between deliberate practice and deliberate play appears likely to be advantageous. Social priming may also be a promising avenue for future research to further our understanding of how creativity is developed. Some programmes for the enhancement of sporting creativity have been recently proposed (e.g. CDF, TCSP) and, although limited, the evidence does support their effectiveness. However, which of the many features of these programmes is responsible for creative development and why remains unclear. Developing this understanding, however, will assist practitioners to implement programmes to develop creativity. Also, while it is recognised that coaches have a pivotal role in the development of sporting creativity, research involving them is still scarce. A potential 'cascade effect' could be investigated, based on the assumption that more creative coaches could develop more creative conditions and, consequently, more creative players. Furthermore, Rasmussen, Østergaard, and Glăveanu (2019) have criticised a perspective of sporting creativity that exclusively emphasises performance, in-game benefits and technical expertise. They propose that creativity should be seen instead as a developmental resource and argue that current performance-oriented visions may lead to overlooking the broader educational benefits that may arise from simply taking part in creative activities, such as increased self-confidence and self-esteem. Along similar lines, Richard and colleagues' (2017) showed that sport diversification and expertise may improve everyday creativity.

Our review found that most of the research conducted, thus far, has employed quantitative and experimental or quasi-experimental designs. Additional insight could be gained from employing other methods such as observation, interviews, or ethnographies to examine *in situ* creativity and its development. This will involve going to the training environment as well as exploring the impact of the broader socio-cultural milieu and personality on sporting creativity. Due to the relative nature of creativity and the importance of domain-specific experts in its understanding and development, such approaches should also engage practitioners (e.g. coaches, scouts) as active participants in all stages of the research process.

Conclusion

This review has demonstrated the lack, as yet, of a widely accepted definition of sporting creativity. We identified important considerations for the conceptualisation of sporting creativity including the distinction between creative thinking (prominent in the research) and creative action, context-specificity, and its emergent nature. The review also demonstrated the influence cognitive conceptualisations of creativity have had on how creativity is assessed, privileging assessment of creative thinking about sport over creative action. Some researchers are beginning to employ ecologically valid assessment (i.e. game-based situations), although these still have limitations often as a result of the definition of creativity that is operationalised. With regards to correlates of creativity, a small range of variables have been examined. Again, the privileging of cognitive definitions has seen attention and pattern recognition being the most commonly researched variables. The review also identified several strategies and programmes that have been proposed for the development of sporting creativity. These show some promise and suggest creativity is trainable, particularly when they include a combination of deliberate practice and deliberate play or less instruction from coaches thereby encouraging greater self-regulated learning. Much remains to be explored and understood about creativity, which presents a range of exciting opportunities for researchers to contribute to this area and further creativity in sport.

Acknowledgements

The authors would like to thank Prof. Robert Hristovski, Prof. Duarte Araujo, Dr. Ilundain-Agurruza, and Maurizio Valenti for their availability and assistance with this review.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Francisco de Sa Fardilha  <http://orcid.org/0000-0003-1679-6579>

Justine B. Allen  <http://orcid.org/0000-0001-9918-9330>

References

- Aggerholm, K., Jespersen, E., & Ronglan, L. T. (2011). Falling for the feint - an existential investigation of a creative performance in high-level football. *Sport, Ethics and Philosophy*, 5(3), 343–358. doi:10.1080/17511321.2011.602589
- Amabile, T. M. (1990). Within you, without you: The social psychology of creativity, and beyond. In M. A. Runco & R. S. Albert (Eds.), *Theories of creativity* (pp. 61–91). Thousand Oaks, CA: SAGE Publications.
- Araújo, D., Hristovski, R., Seifert, L., Carvalho, J., & Davids, K. (2017, December). Ecological cognition: Expert decision-making behaviour in sport. *International Review of Sport and Exercise Psychology*, 9858, 1–25. doi:10.1080/1750984X.2017.1349826
- Arslan, K., Akpunar, F., & Ulucan, K. (2016). Can Neurogulin 1 be an important biomarker for creativity in sports? *Annals of Applied Sport Science*, 4(1), 01–02. doi:10.7508/aass.2016.01.001

- Berends, L., & Johnston, J. (2005). Using multiple coders to enhance qualitative analysis: The case of interviews with consumers of drug treatment. *Addiction Research and Theory*, 13(4), 373–381. doi:10.1080/16066350500102237
- Berry, J., Abernethy, B., & Côté, J. (2008). The contribution of structured activity and deliberate play to the development of expert perceptual and decision-making skill. *Journal of Sport and Exercise Psychology*, 30(6), 685–708. doi:10.1123/jsep.30.6.685
- Boden, M. A. (2004). *The creative mind: Myths and mechanisms* (2nd ed.). New York: Routledge.
- Bowers, M. T., Green, B. C., Hemme, F., & Chalip, L. (2014). Assessing the relationship between youth sport participation settings and creativity in adulthood. *Creativity Research Journal*, 26(3), 314–327. doi:10.1080/10400419.2014.929420
- Braun, V., Clarke, V., & Weate, P. (2017). Using thematic analysis in sport and exercise research. In B. Smith & A. C. Sparkes (Eds.), *Routledge handbook of qualitative research in sport and exercise* (pp. 191–205). New York: Routledge.
- Brown, G. I., & Gaynor, D. (1967). Athletic action as creativity. *The Journal of Creative Behavior*, 1(2), 155–162. doi:10.1002/j.2162-6057.1967.tb00022.x
- Byrge, C., & Hansen, S. (2009). The creative platform: A new paradigm for teaching creativity. *Problems of Education in the 21st Century*, 18, 33–50.
- Byrge, C., & Hansen, S. (2014). *Enhancing creativity for individuals, groups and organizations*. Frederiksberg: Frydenlund Academic.
- Campbell, D. T. (1960, November). Blind variation and selective retentions in creative thought as in other knowledge processes. *Psychological Review*, 67, 380–400. doi:10.1037/h0040373
- Campos, D. (2014). On creativity in sporting activity: With some consequences for education. *FairPlay, Revista de Filosofía, Ética y Derecho del Deporte*, 2(2), 52–80.
- Cavallera, G. M., Boari, G., Labbrozzi, D., & Del Bello, E. D. (2011). Morningness-eveningness personality and creative thinking among young people who play recreational sport. *Social Behavior and Personality: An International Journal*, 39(4), 503–518. doi:10.2224/sbp.2011.39.4.503
- Chow, J. Y. (2013). Nonlinear learning underpinning pedagogy: Evidence, challenges, and implications. *Quest*, 65(4), 469–484. doi:10.1080/00336297.2013.807746
- Costa, I. T., Garganta, J., Greco, P. J., Mesquita, I., & Maia, J. (2011). Sistema de avaliação táctica no Futebol (FUT-SAT): Desenvolvimento e validação preliminar. *Motricidade*, 7(1), 69–84. doi:10.6063/motricidade.7
- Cropley, A. (2006). In praise of convergent thinking. *Creativity Research Journal*, 18(3), 391–404. doi:10.1207/s15326934crj1803_13
- Csikszentmihalyi, M. (1996). *Creativity: Invention, flow and the psychology of discovery and invention*. New York: Harper Collins.
- Davids, K., Güllich, A., Shuttleworth, R., & Araújo, D. (2017). Understanding environmental and task constraints on talent development: Analysis of micro-structure of practice and macro-structure of development histories. In *Routledge handbook of talent identification and development in sport* (pp. 192–206). Abingdon: Routledge.
- Dietrich, A., & Audiffren, M. (2011). The reticular-activating hypofrontality (RAH) model of acute exercise. *Neuroscience and Biobehavioral Reviews*, 35, 1305–1325.
- Dietrich, A., & Haider, H. (2017, January). A neurocognitive framework for human creative thought. *Frontiers in Psychology*, 7, 1–7. doi:10.3389/fpsyg.2016.02078
- Đuriček, M. (1992). Creativity in sports talents: Possibilities and limitations. *Studia Psychologica*, 34(2), 175–182.
- Ericsson, K. A. (2017). Expertise and individual differences: The search for the structure and acquisition of experts' superior performance. *Wiley Interdisciplinary Reviews: Cognitive Science*, 8(1–2), 1–6. doi:10.1002/wcs.1382
- Everhart, B., Kernodle, M., Turner, E., Harshaw, C., & Arnold, D. (1999). Gameplay decisions of university badminton students. *The Journal of Creative Behavior*, 33(2), 138–149. doi:10.1002/j.2162-6057.1999.tb01043.x
- Finke, R. A., Ward, T. B., & Smith, S. M. (1992). *Creative cognition: Theory, research, and applications*. Cambridge, MA: The MIT Press.

- Furley, P., & Memmert, D. (2015, February). Creativity and working memory capacity in sports: Working memory capacity is not a limiting factor in creative decision making amongst skilled performers. *Frontiers in Psychology*, 6, 1–7.
- Furley, P., & Memmert, D. (2018). Can creative role models prime creativity in soccer players? *Psychology of Sport and Exercise*, 37, 1–9. doi:10.1016/j.psychsport.2018.03.007
- Furley, P., Memmert, D., & Heller, C. (2010). The dark side of visual awareness in sport: Inattention blindness in a real-world basketball task. *Attention, Perception & Psychophysics*, 72(5), 1327–1337. doi:10.3758/APP.72.5.1327
- Gardner, H. (1993). *Creating minds: An anatomy of creativity seen through the lives of Freud, Einstein, Picasso, Stravinsky, Eliot, Graham, and Gandhi*. New York: Basic Books.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston: Houghton Mifflin.
- Grant, M. J., & Booth, A. (2009). A typology of reviews: An analysis of 14 review types and associated methodologies. *Health Information and Libraries Journal*, 26, 91–108. doi:10.1111/j.1471-1842.2009.00848.x
- Greco, P., Memmert, D., & Morales, J. C. P. (2010). The effect of deliberate play on tactical performance in Basketball. *Perceptual and Motor Skills*, 110(3), 849–856. doi:10.2466/pms.110.3.849-856
- Greenhalgh, T., & Peacock, R. (2005). Effectiveness and efficiency of search methods in systematic reviews of complex evidence: Audit of primary sources. *BMJ (Clinical Research Ed.)*, 331(7524), 1064–1065. doi:10.1136/bmj.38636.593461.68
- Guilford, J. P. (1967). *The nature of human intelligence*. New York: McGraw-Hill.
- Harrison, C. M. (2016, February). Bebop on the Hockey Pitch: Cross-disciplinary creativity and skills transfer. *Frontiers in Psychology*, 7, 1–5. doi:10.3389/fpsyg.2016.00123
- Hendry, D. T., Williams, A. M., & Hodges, N. J. (2018). Coach ratings of skills and their relations to practice, play and successful transitions from youth-elite to adult-professional status in soccer. *Journal of Sports Sciences*, 36(17), 2009–2017.
- Hopsicker, P. (2011). In search of the 'sporting genius': Exploring the benchmarks to creative behavior in sporting activity. *Journal of the Philosophy of Sport*, 38(1), 113–127. doi:10.1080/00948705.2011.9714553
- Hristovski, R., Davids, K., Araújo, D., & Passos, P. (2011). Constraints-induced emergence of functional novelty in complex neurobiological systems: A basis for creativity in sport. *Psychology and Life Sciences*, 15(2), 175–206.
- Hristovski, R., Davids, K., Passos, P., & Araújo, D. (2012). Sport performance as a domain of creative problem solving for self-organizing performer-environment systems. *The Open Sports Sciences Journal*, 5(1), 26–35. doi:10.2174/1875399X01205010026
- Hüttermann, S., Nerb, J., & Memmert, D. (2018). The role of regulatory focus and expectation on creative decision making. *Human Movement Science*, 62, 169–175. doi:10.1016/j.humov.2018.10.006
- Igorov, M., Predoiu, R., Predoiu, A., & Igorov, A. (2016). *Creativity, resistance to mental fatigue and coping strategies in junior women handball players*. 5th international congress of physical education, sports and kinethotherapy, Bucharest, 2015.
- Ilundáin-Agurruza, J. (2017). Muscular imaginings—A phenomenological and enactive model for imagination. *Sport, Ethics and Philosophy*, 11(1), 92–108. doi:10.1080/17511321.2017.1294197
- Kaufman, J., & Sternberg, R. (2007). Creativity. *Change: The Magazine of Higher Learning*, 39(4), 55–60. doi:10.3200/CHNG.39.4.55-C4
- Kempe, M., & Memmert, D. (2018). "Good, better, creative": The influence of creativity on goal scoring in elite soccer. *Journal of Sports Sciences*, 36(21), 2419–2423. doi:10.1080/02640414.2018.1459153
- Kim, K. H. (2011). The creativity crisis: The decrease in creative thinking scores on the torrance tests of creative thinking. *Creativity Research Journal*, 23(4), 285–295. doi:10.1080/10400419.2011.627805
- Krampen, G., Freiling, J., & Willems, L. (1996). *Kreativitätstest für Vorschul- und Schulkinder. Version für die psychologische Anwendungspraxis (KVS-P)*. [Creativity test for preschool and school children (KVSP): Version for preschool and school children]. Göttingen: Hogrefe.
- Krein, K., & Ilundáin-Agurruza, J. (2017). High-level enactive and embodied cognition in expert sport performance. *Sport, Ethics and Philosophy*, 11(3), 370–384. doi:10.1080/17511321.2017.1334004

- Lacerda, T., & Mumford, S. (2012). The genius in art and in sport: A contribution to the investigation of aesthetics of sport. *Journal of the Philosophy of Sport*, 37(2), 182–193. doi:10.1080/00948705.2010.9714775
- Lauder, A., & Piltz, W. (2013). *Play practice* (2nd ed.). Champaign: Human Kinetics.
- Leso, G., Dias, G., Ferreira, J., Gama, J., & Couceiro, M. S. (2017). Perception of creativity and game intelligence in soccer. *Creativity Research Journal*, 29(2), 182–187. doi:10.1080/10400419.2017.1302779
- Martin, J., & Cox, D. (2016). Positioning Steve Nash: A theory-driven, social psychological, and biographical case study of creativity in sport. *The Sport Psychologist*, 30(4), 388–398. doi:10.1123/tsp.2016-0002
- Mednick, S. (1962). The associative basis of the creative process. *Psychological Review*, 69(3), 220–232. doi:10.1037/h0048850
- Memmert, D. (2006). Developing creative thinking in a gifted sport enrichment program and the crucial role of attention processes. *High Ability Studies*, 17(1), 101–115. doi:10.1080/13598130600947176
- Memmert, D. (2007). Can creativity be improved by an attention-broadening training program? An exploratory study focusing on team sports. *Creativity Research Journal*, 19(2-3), 281–291. doi:10.1080/10400410701397420
- Memmert, D. (2010). Testing of tactical performance in youth elite soccer. *Journal of Sports Science and Medicine*, 9(2), 199–205.
- Memmert, D. (2011). Creativity, expertise, and attention: Exploring their development and their relationships. *Journal of Sports Sciences*, 29(1), 93–102. doi:10.1080/02640414.2010.528014
- Memmert, D. (2015). *Teaching tactical creativity in sport: Research and practice*. Routledge studies in physical education and youth sport. Abingdon: Routledge.
- Memmert, D. (2017). Tactical creativity in sport. In J. C. Kaufman, V. P. Glăveanu, & J. Baer (Eds.), *The Cambridge handbook of creativity across domains* (pp. 479–491). Cambridge: Cambridge University Press.
- Memmert, D., Baker, J., & Bertsch, C. (2010). Play and practice in the development of sport-specific creativity in team ball sports. *High Ability Studies*, 21(1), 3–18. doi:10.1080/13598139.2010.488083
- Memmert, D., & Furley, P. (2007). “I spy with my little eye!”: Breadth of attention, inattention blindness, and tactical decision making in team sports. *Journal of Sport & Exercise Psychology*, 29(3), 365–381. doi:10.1123/jsep.29.3.365
- Memmert, D., Hüttermann, S., & Orliczek, J. (2013). Decide like Lionel Messi! The impact of regulatory focus on divergent thinking in sports. *Journal of Applied Social Psychology*, 43(10), 2163–2167. doi:10.1111/jasp.12159
- Memmert, D., & Perl, J. (2009a). Game creativity analysis using neural networks. *Journal of Sports Sciences*, 27(2), 139–149. doi:10.1080/02640410802442007
- Memmert, D., & Perl, J. (2009b). Analysis and simulation of creativity learning by means of artificial neural networks. *Human Movement Science*, 28(2), 263–282. doi:10.1016/j.humov.2008.07.006
- Memmert, D., & Roth, K. (2003). Individualtaktische Leistungs- diagnostik im Sportspiel [Diagnostics of individual tactical performance in sports games]. *Spektrum Der Sportwissenschaft* [Spectrum of Sport Science], 15, 44–70.
- Memmert, D., & Roth, K. (2007). The effects of non-specific and specific concepts on tactical creativity in team ball sports. *Journal of Sports Sciences*, 25(12), 1423–1432. doi:10.1080/02640410601129755
- Moraru, A., Memmert, D., & van der Kamp, J. (2016). Motor creativity: The roles of attention breadth and working memory in a divergent doing task. *Journal of Cognitive Psychology*, 28(7), 856–867. doi:10.1080/20445911.2016.1201084
- Oh, J., Joung, K., Kim, H.-K., Choi, H., Kim, N., & Sung, J. (2010). *Coaches' views on the development of creativity of Korean football*. Research Consortium Conference, Indianapolis, March 2010.
- Orth, D., van der Kamp, G. J. P., Memmert, D., & Savelsbergh, G. J. P. (2017, October). Creative motor actions as emerging from movement variability. *Frontiers in Psychology*, 8, 1–8.
- Owen, A., Twist, C., & Ford, P. (2004). Small-sided games: The physiological and technical effect of altering pitch size and player numbers. *Insight*, 7, 50–53.

- Rasmussen, L. J. T., & Østergaard, L. D. (2016). The creative soccer platform: New strategies for stimulating creativity in organized youth soccer practice. *Journal of Physical Education, Recreation & Dance*, 87(7), 9–19. doi:10.1080/07303084.2016.1202799
- Rasmussen, L. J. T., Østergaard, L. D., & Glăveanu, V. P. (2019). Creativity as a developmental resource in sport training activities. *Sport, Education and Society*, 24, 491–506.
- Richard, V., Abdulla, A. M., & Runco, M. A. (2017). Influence of skill level, experience, hours of training, and other sport participation on the creativity of elite athletes. *Journal of Genius and Eminence*, 2(1), 65–76. doi:10.18536/jge.2017.04.02.01.07
- Roca, A., Ford, P. R., & Memmert, D. (2018). Creative decision making and visual search behavior in skilled soccer players. *PLoS ONE*, 13(7), e0199381. doi:10.1371/journal.pone.0199381
- Roco, M. (2004). *Creativitate și inteligență emoțională*. Iaș: Editura Polirom.
- Runco, M. (2014). *Creativity: Theories and themes: Research, development, and practice* (2nd ed.). Amsterdam: Academic Press.
- Runco, M. A., & Chand, I. (1994). Problem finding, evaluative thinking, and creativity. In M. A. Runco (Ed.), *Problem finding, problem solving, and creativity* (pp. 40–76). Norwood, NJ: Ablex.
- Runco, M. A., & Jaeger, G. J. (2012). The standard definition of creativity. *Creativity Research Journal*, 24(1), 92–96. doi:10.1080/10400419.2012.650092
- Santos, S., Coutinho, D., Gonçalves, B., Schöllhorn, W., Sampaio, J., & Leite, N. (2018). Differential learning as a key training approach to improve creative and tactical behavior in soccer. *Research Quarterly for Exercise and Sport*, 89(1), 11–24.
- Santos, S., Jiménez, S., Sampaio, J., & Leite, N. (2017). Effects of the Skills4Genius sports-based training program in creative behavior. *PLoS ONE*, 12(2), 1–7. doi:10.1371/journal.pone.0172520
- Santos, S. D. L., Memmert, D., Sampaio, J., & Leite, N. (2016). The spawns of creative behavior in team sports: A creativity developmental framework. *Frontiers in Psychology*, 7, 1–14. doi:10.3389/fpsyg.2016.01282
- Schöllhorn, W. I., Mayer-Kress, G., Newell, K. M., & Michelbrink, M. (2009). Time scales of adaptive behavior and motor learning in the presence of stochastic perturbations. *Human Movement Science*, 28(3), 319–333. doi:10.1016/j.humov.2008.10.005
- Simonton, D. K. (2012). Taking the U.S. patent office criteria seriously: A quantitative three-criterion creativity definition and its implications. *Creativity Research Journal*, 24(2–3), 97–106. doi:10.1080/10400419.2012.676974
- Smith, W. (2014). Fundamental movement skills and fundamental games skills are complementary pairs and should be taught in complementary ways at all stages of skill development. *Sport, Education and Society*, 21(3), 431–442. doi:10.1080/13573322.2014.927757
- Sternberg, R., & Lubart, T. (1999). The concept of creativity: Prospects and paradigms. In R. Sternberg (Ed.), *Handbook of creativity* (pp. 3–15). Cambridge: Cambridge University Press.
- Sternberg, R. J., & Lubart, T. I. (1991). An investment theory of creativity and its development. *Human Development*, 34, 1–31. doi:10.1159/000277029
- Tamarit, X. (2016). *Periodizacion tactica versus Periodizacion tactica*. Buenos Aires: Librofutbol.com.
- Tan, C. W. K., Chow, J. Y., & Davids, K. (2012). 'How does TGfU work?': examining the relationship between learning design in TGfU and a nonlinear pedagogy. *Physical Education & Sport Pedagogy*, 17(4), 331–348. doi:10.1080/17408989.2011.582486
- Tanggaard, L., Laursen, D. N., & Szulevics, T. (2016). The grip on the handball – a qualitative analysis of the influence of materiality on creativity in sport. *Qualitative Research in Sport, Exercise and Health*, 8(1), 79–94. doi:10.1080/2159676X.2015.1012546
- Torrents, C., Ric, A., Hristovski, R., Torres-Ronda, L., Vicente, E., & Sampaio, J. (2016). Emergence of exploratory, technical and tactical behavior in small-sided soccer games when manipulating the number of teammates and opponents. *PLoS ONE*, 11(12), 1–15. doi:10.1371/journal.pone.0168866
- Veraksa, A. N., & Gorovaya, A. E. (2011). Effect of imagination on sport achievements of novice soccer players. *Psychology in Russia: State of Art*, 5(1), 495–504. doi:10.11621/pir.2011.0032
- Wallas, G. (1926). *The art of thought*. New York: Harcourt, Brace and Company.
- Yamamoto, Y., & Yokoyama, K. (2011). Common and unique network dynamics in football games. *PLoS ONE*, 6(12), 1–6. doi:10.1371/journal.pone.0029638